

1.1.1. Drogue Parachute (Ballute)

A ballute is a type of parachute that takes the form of a balloon, acting as a low-density high speed decelerator. Described as a combination between a parachute and a balloon, they are typically used at high altitudes and speeds for their higher stability, at the cost of a shape that provides a lower drag coefficient.

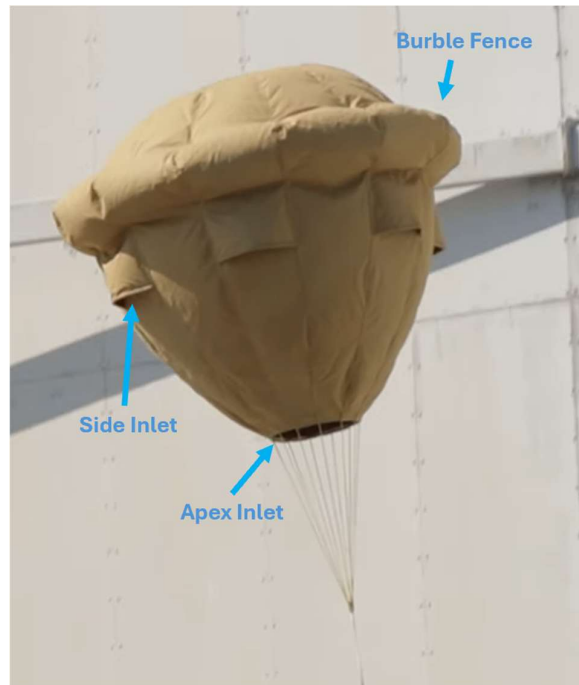


Fig. 188: Copenhagen Suborbitals Ballute

The ballute is not alone in the low-density decelerator space. Disk-band-gap (DBG) parachutes are also commonly used for high altitude descent control. They have the advantage of a higher drag coefficient, allowing for a lighter design for the same descent requirements. However their long shroud lines and geometry makes them susceptible to tangling. With no clear method to confidently verify inflation stability, the ballute is inherently a safer device to use for high altitude descent control for its tangling resistance.

1.1.1.1. Initial Sizing

To begin designing the ballute, first an estimated coefficient of drag for a 4 foot, 80 degree toroidal burble fence design will be used. The average of 29 drop tests from the Arcas sounding rocket recovery system [29] resulted in a $C_d = 1.048$.

Applying a factor of safety of 1.125 will ensure speeds are within allowable limits during the ballute descent. Hence, the ballute is required to slow the rocket to a descent rate of 40 m/s at 30,000 ft. Sizing calculations shown in appendix 5.6.6.1 determine a ballute diameter of 1.465 m resulting in a descent speed of 24 m/s at 1000 ft.

1.1.1.2. Geometry

The toroidal fence ballute was selected for its simplicity in design, inherent tangling resistance and hand manufacturability as demonstrated by Copenhagen Suborbitals [30]. The ballute angle and burble fence ratios will be derived from the Arcas program seen in Figure 191.

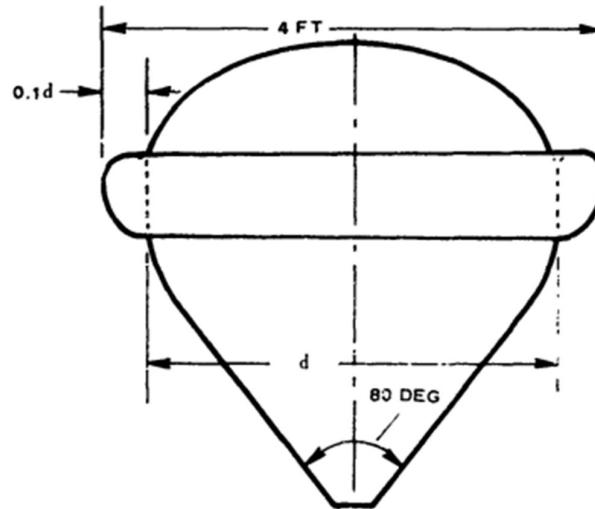


Fig. 189: Arcas Program Ballute Geometry

12 gores will be chosen to match Copenhagen suborbitals design and allow for 6 side air inlets. Ballute geometry generally follows an isotenoid shape, meaning that fabric stresses are uniformly distributed throughout the canopy. The derivation of such shapes is complex and outside the scope of this design. To account for this, general shape attributes will be derived from similar isotenoid designs. As demonstrated further in the report, the pressures experienced and the flight regime are orders of magnitude less than similar designs so a representative factor of safety is deemed acceptable.

Based on the drag estimate used from the Arcas rocket development. An 80 degree 10% diameter burble fence design will be used. This will inform the calculation of surface area and volume.

In-line with the methods from Goodyear Aerospace [31], the surface area and volume of a ballute main body is calculated based on corrections applied to a sphere of representative diameter.

$$A_B = 4.35\pi R^2 = 4.35\pi(0.7325)^2 = 7.3330 \text{ m}^2$$

$$V_B = \frac{14}{9}\pi R^3 = \frac{14}{9}\pi(0.7325)^3 = 1.9207 \text{ m}^3$$

The area and volume 10 % burble fence on the equator of the ballute is derived geometrically as;

$$A_{bf} = 1.42\pi R^2 = 1.42\pi(0.7325)^2 = 2.3936 \text{ m}^2$$

$$V_{bf} = 0.1364\pi R^3 = 0.1364\pi(0.7325)^3 = 0.1684 \text{ m}^3$$

1.1.1.3. Inlet Sizing

With a fully inflated 1.465 m ballute, the desired descent speeds can be obtained. The next step of design is to size the apex and side inlets appropriately to ensure the ballute inflates properly and maintains its shape. Seen in Figure 192 below, the atmospheric density at 100,000 ft is 73 times less than sea level density. This creates a difficult environment for a ballute to inflate in.

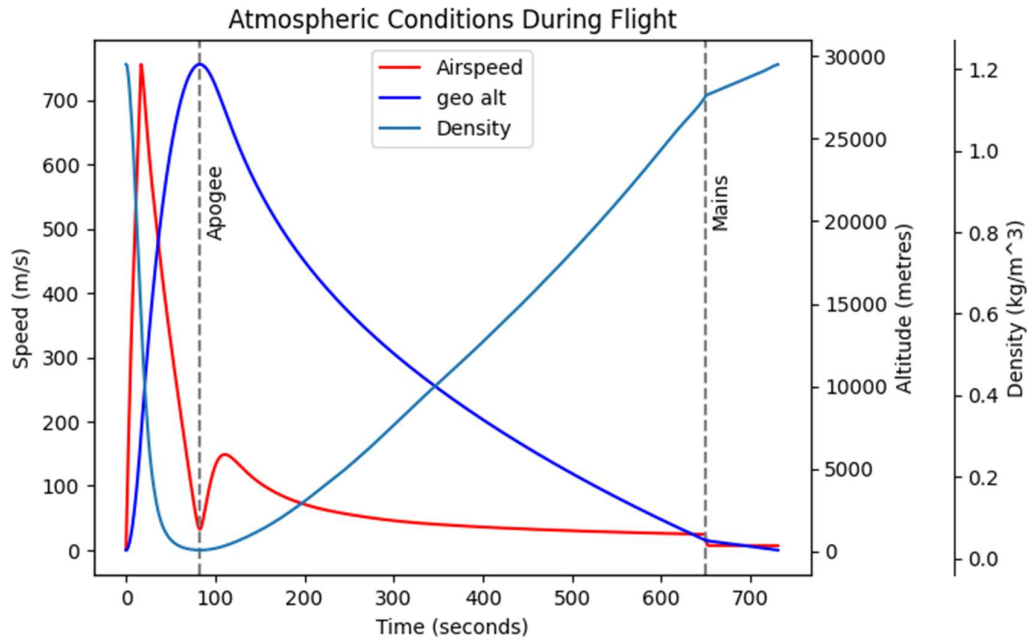


Fig. 190: Flight Profile Conditions over Time

Another consideration is the available ram pressure during descent. This can be derived via an isentropic stagnation pressure seen in the following equation.

$$\frac{p_0}{p} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma - 1}}$$

Analysing this stagnation pressure over the entire flight yields the results in Figure 193 which shows a positive pressure inside the ballute is achievable through ram air inlets throughout the entire flight. The maximum pressure differential is just after apogee where velocity locally maximises in the thin air before slowing down at a lower altitude.

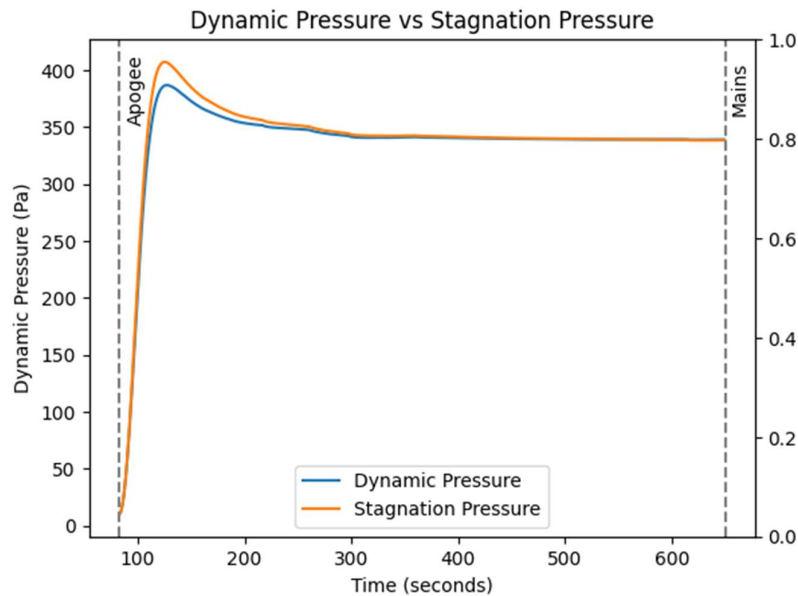


Fig. 192: Dynamic and Stagnation Pressure Difference over flight

As seen in Figure 192, the stagnation pressure ratio remains around 1 which is adequate for full inflation when factoring the varied pressure field around a ballute during descent [32].

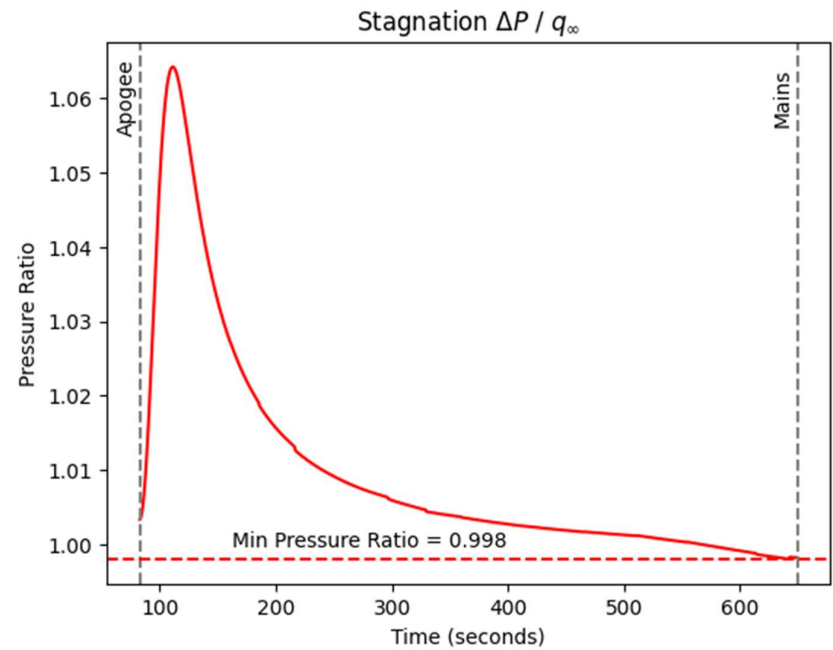


Fig.193: Stagnation-to-Dynamic Pressure ratio over Flight

Using this information a pressure differential can be estimated for different parts of the flight to inform ideal mass flowrate. For both the minimum inflation pressure differential and apogee a total inlet size can be determined through the compressible massflow formula seen in *Equation 1* for an inflation time of 3 seconds and an inlet discharge coefficient of 0.5, inline with a sharp circular inlet. This assumes an impermeable ballute, which is an acceptable assumption given ripstop nylon and other weaves can have extremely low porosity, with seams being able to be taped and fabrics having treatments such as silicon sealing. The method to arrive at the inlet sizing is detailed in the [appendix](#) 5.6.6.2 are shown in Table 46.

$$\dot{m} = \frac{Ap_t}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R}} M \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\frac{\gamma + 1}{2(\gamma - 1)}}$$

Equation 1: Compressible massflow formula

Table 46: Required Inlet Area for Max-Q and Mains Deployment

Altitude (m)	$P_0 - q$ (Pa)	Air (kg)	Required inlet area (cm^2)
30,000	338	0.042	434
660	10.5	2.326	566

With this inlet area and a FoS of 2 to account for fabric flapping affecting mass flowrate inside the ballute, side inlets and apex dimensions can be determined. With an arbitrary ratio of 0.5, the apex diameter must be 25 cm, and the side inlet must each have an area of at least 1 cm². This inlet to diameter ratio is inline with Copenhagen suborbitals ballute ratio, although the remaining side inlet area is quite small. The assumptions leading to this number do not account for fabric deformation altering flowrate. Hence side inlet ratios from similar designs will be used. To view the code used to generate the graphs and results from this section, see *appendix 5.6.6.2*.

1.1.1.4. Material Selection

The ballute will be deployed at 100,000 ft and be the decelerator for the majority of the recovery phase. For this it needs to be durable enough to survive the stresses and conditions while falling through the atmosphere. For this it needs to;

- Withstand temperatures without compromising strength
- Withstand stresses of inflation without bursting
- Withstand flagging forces during descent

Of these three conditions, the flagging forces during inflation are the highest. The Arcas Rocketsondes Ballute program estimated through testing that the flagging force in the fabric peaks at $2qR$. Where R is the equatorial radius of the ballute and q is the dynamic pressure of deployment. For a conservative estimate, the peak dynamic pressure ~45 seconds after apogee will be used to drive this design constraint.

$$2qR = 2(386.790)(0.7325) = 566.629 \text{ N/m}$$

This maximum predicted stress will inform material choice and will be further validated by physical testing. It is important to note at this phase the Ballute will be inflated, with the internal pressure dampening the flagging forces.

Another design requirement for the ballute canopy is a low porosity material. This is important to aid in canopy rigidity during descent. The previous inlet sizing assumed mass lost through porosity was negligible due to favourable porosity characteristics of fabrics such as woven nylon which is anticipated for use.

Before selecting a fabric a temperature profile must be calculated to ensure there are no adverse effects on fabric strength that need to be considered. To calculate this, total temperature will be analysed throughout the flight profile.

$$\frac{T}{T_T} = (1 + \frac{\gamma - 1}{2} M^2)^{-1}$$

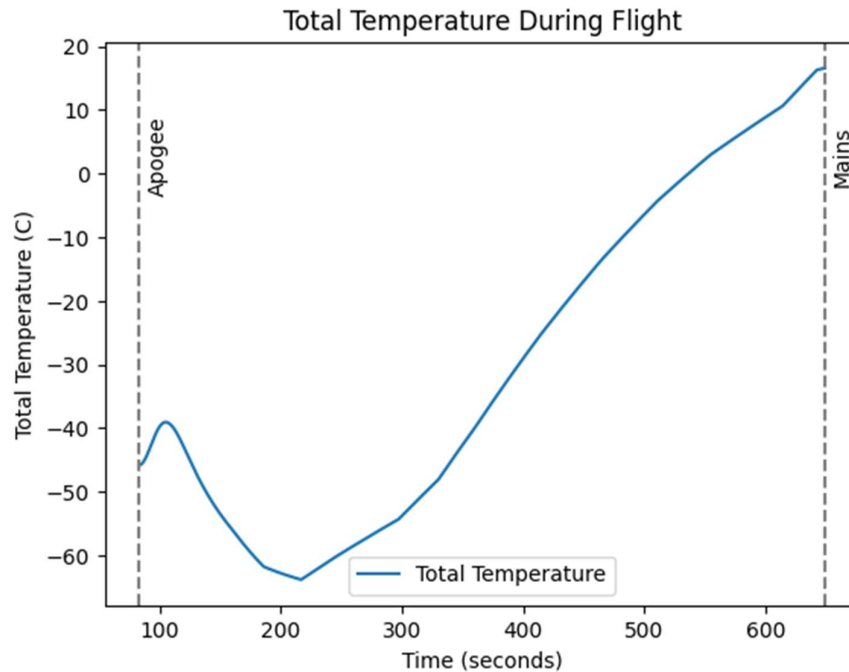


Fig. 194: Total Temperature of Air throughout Flight

As seen in Figure 195 the minimum temperature of stagnated air during flight is -63.9°C and occurs at an altitude of 16450 m.

These impose certain design criteria that will drive the selection of Ballute material.

- The capacity to withstand the environment without degradation
- The highest strength to withstand flagging forces for the least fabric weight
- Low porosity
- Manufacturability

The main fibres that will be considered for the ballute are nylon or polyester. These are organic and synthetic fibers that are commonly used in aerospace applications. These base fibers as textiles can also be treated/coated to tailor them towards more specific applications.

Additionally, the stress behaviour of these fibers can be observed from testing on polyester and nylon based textiles by the Ottawa Defence Research Establishment's paper on the effect of low temperatures on coated and uncoated fabrics [33]. This largely shows that reduction in temperature as low as -40°C on all nylon and polyester blends results in a 30% increase in breaking load and a decrease in elongation (stiffening), which is a promising result for use on the ballute. Polyurethane and neoprene coatings performed well, with the only detrimental degradation being the PVC coated nylon scrims which were deemed unusable below 0°C . Taking this into account with the previously calculated flagging stress a factor of safety of at least 5 will be chosen, initial candidates are summarised in Table 47.

Table 47: Ballute materials summary

Fabric	Weight (gsm)	UTS (weft/warp)(N) _{AS_2001.2.3.2}	FoS	Coating
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70D Diamond Nylon-R	88	5490/4020	7.1	Silicon/PU
50D Polyester-R	85	5670/3350	5.9	Silicon/PU

Aramid fibres such as Kevlar and Nomex were analysed but found to have a high fabric weight, high porosity and excessive factor-of-safety. For these reasons aramids will not be considered for the ballute application.

Both the nylon and polyester performed well relative to their room temperature samples in the Ottawa Defence experiment with the two selected fabrics having adequately high FoS. They did show signs of hardening with a higher strength and lower elongation, however with the FoS displayed, the increase in rigidity is not a concern.

Both materials have zero porosity inline with ASTM D0737. With identical coatings, this makes both fabrics strong candidates. Polyester will be chosen as the candidate material due to nylon's susceptibility to water absorption. This can reduce the tensile strength of the woven fabric significantly [34] which can pose additional risk in a sub-zero environment, as the freezing of trapped moisture can form crystals in the nylon fibre which may lead to catastrophic fabric degradation.

Polyester is inherently hydrophobic and mitigates this issue significantly which will simplify ground handling requirements of the constructed ballute. The use of 50D Ripstop Polyester will result in an approximate total ballute fabric weight of 827 grams.

1.1.1.5. Shroud Lines and Bridle Design

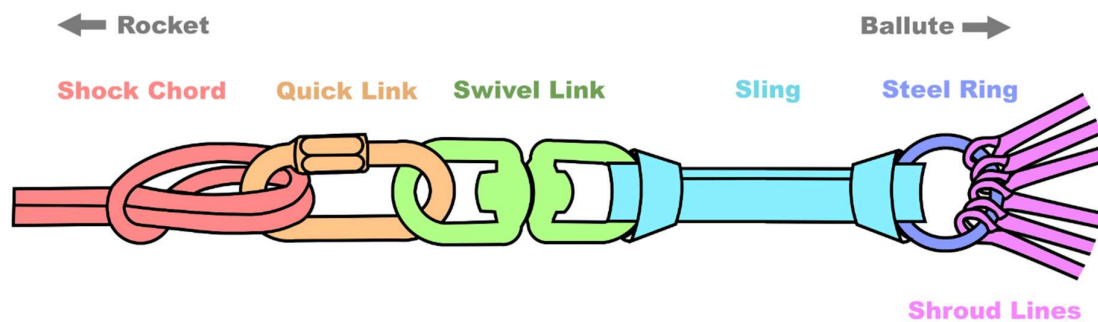


Fig. 195: Ballute attachment hardware

In addition to the design of the ballute canopy, shroud lines and the bridle must be designed to attach the ballute to the entire recovery system. The system is detailed in Figure 196, starting from the shroud lines, working down to the shock chord.

For the shroud lines, for proven use and manufacturability in parachute systems, Spectra chords will be chosen [35]. Honeywell's Spectra braided line is made from ultra-high-molecular-weight (UHMW) polyethylene fibre which has high tensile strength, abrasion resistance, and importantly a sufficiently low ductile-to-brittle transition temperature of -70 degrees which outperforms other synthetic fibres such as nylon [36]. The maximum steady state drag force experienced by the ballute is 683 newtons which occurs 44.5 seconds after launch. With 12 gores this means each shroud line will experience a small 56 newtons. This clearly highlights the priority of the design on environmental survivability for this component. Spectra far exceeds these loading requirements and will perform well in the low temperatures expected during flight. The specific spectra line that will be used is hollow braided 2 mm spectra line with a breaking strength of 3,924 newtons.

This force requirement also applies to the bridle that connects the ballute to the larger recovery system. This will be made from polyethylene much like the shroud lines. The specific product is a dyneema climbing sling with a breaking force of 22 kN, much higher than required. One side of the sling will attach to the swivel link, the other to a pair of welded steel rings that will connect the bridle to the shroud lines. The steel rings used are 25 mm loop 316 stainless steel with a breaking load limit of 19.5 kN.

The swivel link used will be a 13 kN stainless steel swivel link that will attach through a 5.3 kN ¼" stainless steel quick link that will tie into a loop in the shroud line.

1.1.1.6. Initial CFD Analysis

Initial CFD analysis was conducted on the ballute to characterise its coefficient of drag. This was completed with the assumption that the ballute is a rigid object, and no internal air flow was modelled. This involved running one scenario where the Cd was calculated with the ballute exposed to the air stream at 90,000 ft with the theoretical speed achieved at such altitude compared to the air density and theoretical speed at 2,000 ft. Both simulation scenarios have resulted in a Cd of 0.95, proving the reliability of the CFD model.

Additionally, an iterative approach was used to converge on a more accurate drag coefficient estimate for the ballute by coupling the theoretical values provided by OpenRocket to values provided by the CFD analysis. This test found the final drag coefficient to be 1.035 which is within 2% of the predicted Cd of 1.05 based on theoretical calculations. Further details outlining the simulation methodology and results are detailed in [appendix 5.2.4.8](#)

1.1.1.7. Further Validation

To further validate the theoretical approximation of ballute behaviour previously outlined, static CFD will be conducted to observe a more detailed pressure field around the ballute during critical points of descent as detailed in [appendix 5.5.5.6](#). Further simulations can be completed by modelling the inflation and increase in internal pressure of the ballute.

Additionally, post-manufacturing testing procedures can be employed to validate the construction and performance

1.1.1.8. Manufacturing and Assembly

The construction of the ballute poses a difficult sewing challenge to construct the canopy, and attach the periphery hardware.

As executed by Copenhagen suborbitals, the ballute will be made in two halves and then joined at the longitudinal vents. The bottom gores will form the burble fence when sewn to the top half at two rings around the equator. Alternating bottom gores will include a vent hole in their upper section which will be covered by a square vent shroud that will direct air into the ballute. The burble fence will be inflated by airflow through partially unsewn seams of the top gores at the equator. Flat felled seam stitches will be used with high strength polyester thread where possible, with flat seams being used elsewhere. For more detail on this process, please refer to [appendix 5.2.5.4](#).

To assemble the ballute attachment hardware, start from the shock chord and move to the right. It may be useful to revisit Figure 196. On the shock chord leading from the upper to the lower section of the rocket relative to the drogue bay, tie a loop in the chord and attach the quick link with the swivel attached. Tie a girth-hitch onto the other end of the swivel with the dyneema webbing loop. Tie this dyneema webbing to 2 welded steel rings with another girth-hitch. To tie the shroud lines to the steel rings, they will be split, 6 per side onto each ring. The hollow weave spectra lines will be attached to the steel ring via a long bury splice (finger trap splice) seen in Figure 197. After the splice will be sewn together for increased strength and stability. This attachment is deemed suitable due to the high FoS of the braided line and the high efficiency of this nature of splice [37].



Fig. 196: Long bury splice (finger trap splice) loop

1.1.1.9. Testing Plans

Ground Inflation Testing

In order to physically validate its strength, the ballute will be subject to on-ground physical testing prior to launch. One such ground test will be an inflation car-tow test, which will aim to simulate on the ground the inflation forces experienced by the ballute at 100,000ft.

From Fig 193. *Dynamic and Stagnation Pressure Differences over Flight*, it can be observed that pressure will gradually increase from 0Pa to 410Pa, occurring between 30,000m and 23,000m. The change in pressure over this time period follows a linear trend, hence we will take $\frac{dq}{dt} = \frac{410Pa}{30} = 13.7 Pa/s$.

Our aim is to best create these conditions at ground level. We can start by calculating the ground speed required to achieve this pressure:

$$\text{At sea level, dynamic pressure} = \frac{1}{2}\rho v^2$$

$$\text{Therefore: } v = \sqrt{\frac{2P}{\rho}} = \sqrt{\frac{2(410)}{1.225}} = 25.9m/s = 93.25km/h$$

Therefore, to most accurately represent the inflation loading at 100,000ft. The car must tow the parachute from 0 to 93.25km/h over a 30s period, steadily accelerating. While this won't perfectly simulate the forces experienced at 100,000ft, it will be a valuable representation.

It is also recommended that tests up to 100km/h are performed, to observe the parachute at pressures greater than expected (470 Pa at 100km/h). The testing procedure can be found in the [appendix 5.2.5.1](#).

Seam Testing

Seams are common points of failure on all parachutes. This is because of joint efficiency, which is the loss of strength occurring at the connection between two different materials. Hence, we will use tensile testing to determine the strength of seams. It should be noted that this requirement also informed the selection of the parachute canopy material. Assuming a joint efficiency of 50%, a material tensile strength of 4000N/m will be required to achieve a FoS of 3 at the Seams.

Tensile testing will be conducted on samples of seams at the prototyping phase, where they must have a FoS greater than 3 in relation to the maximum flagging force experienced by the chute (566.92 N/m). If this stage is passed, ballute manufacturing can go ahead.

Tensile testing will also be conducted on samples of seams completed during the manufacturing phase. These seams must also all pass with a FoS of 3 to validate the Ballute's seam strength.